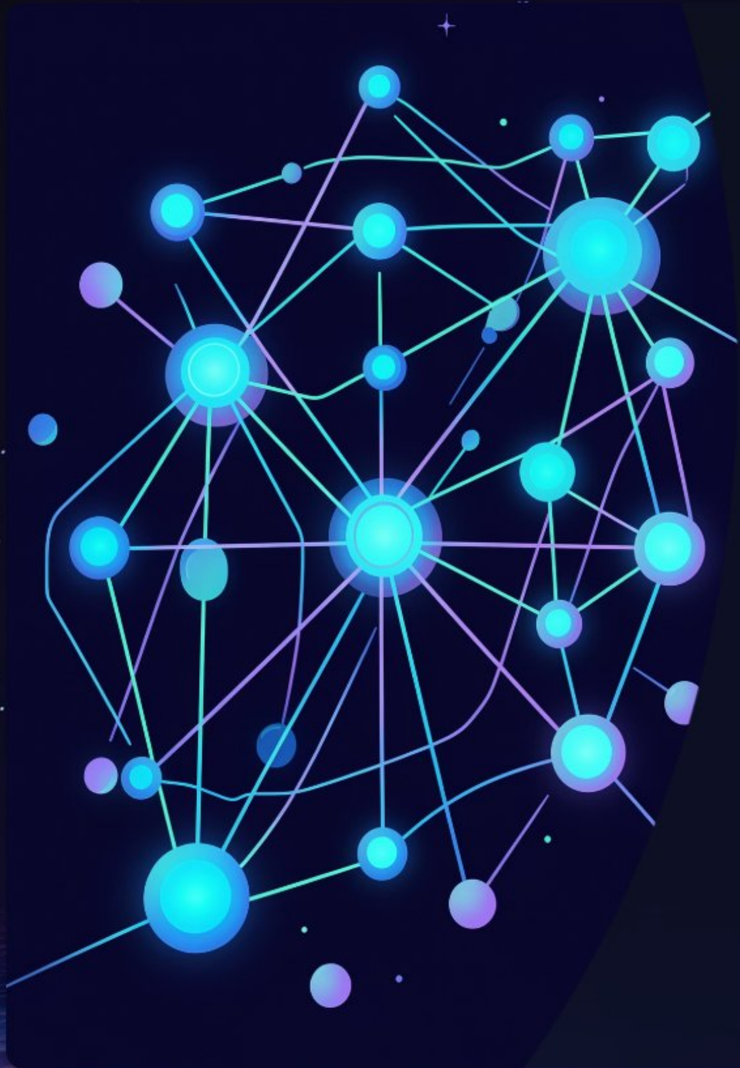

Real-Time News Classification and Summarisation using ML, DL & LLMs

A final Year B.Tech Project Presentation by:

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Presentation Outline

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- Dataset Description
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- Data Preprocessing and Feature Extraction
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Introduction

The Information Overload

• Millions of News articles are generated every day across multiple domains online.

• Manually organising and understanding this information is time-consuming and inefficient.

• News classification helps in automatically categorising articles into predefined topics.

• Summarization further enables users to quickly understand key information.

• Machine Learning, Deep Learning, and LLMs provide effective solutions for automating these tasks.

📰 Our Goal?

Automatically assigning news articles to predefined categories using ML, DL, and LLM techniques.

🎯 Why It Matters

Reduces human effort, improves accuracy, and enables real-time intelligent content filtering.

Problem Statement & Objectives

The Core Problem

- The rapid growth of online news makes manual classification difficult and inefficient.
- Users often struggle to find relevant news due to lack of proper categorisation.
- Reading full articles is time-consuming given the large volume of content.

⚠ Currently, there is no widely adopted system that integrates both news classification and summarization into a single unified framework.

Most existing approaches focus on static classification tasks using predefined datasets, while summarization is often treated as a separate process.

This lack of integration limits the efficiency and real-world applicability of such systems.

01

Build a robust classification-summarization pipeline

Spanning classical ML to advanced DL models.

02

Compare model performance rigorously

Across accuracy, precision, recall, and F1-score.

03

Integrate LLM-based summarisation

To generate concise, human-readable article summaries.

04

Deploy a real-time classification interface

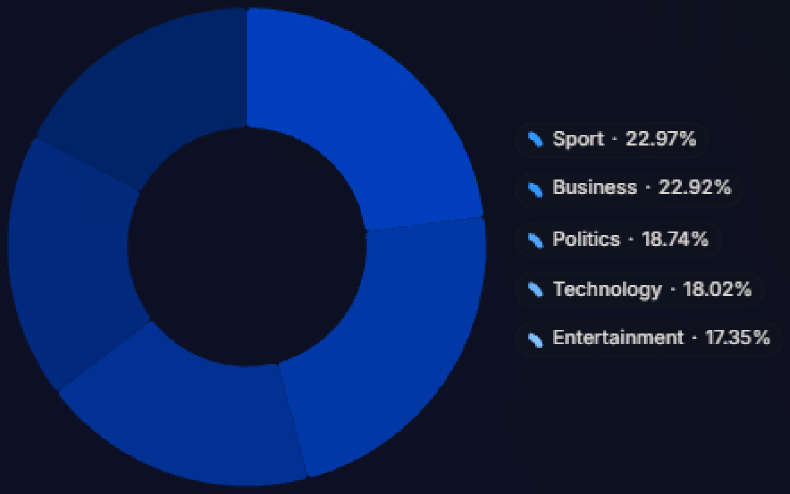
For live news input and instant category prediction.

Dataset Description

BBC News Dataset

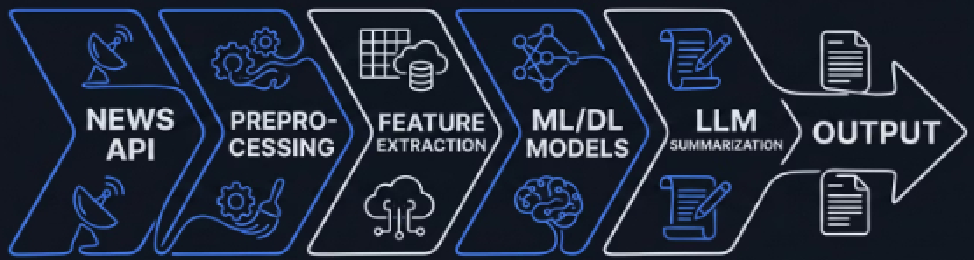
- A widely benchmarked corpus of **2,225 news articles** sourced from the BBC website and Kaggle
- It spans five distinct topic categories. Each article is labelled and suitable for supervised NLP tasks.
- Data is balanced across categories.

Category	Articles
Politics	417
Sport	511
Business	510
Entertainment	386
Technology	401



Methodology - End-to-End Pipeline

Our comprehensive pipeline integrates various stages to transform raw news data into classified and summarized outputs, ensuring efficiency and accuracy.



- Real-time news articles are fetched using APIs
- Text is preprocessed (cleaning, tokenization, lemmatization)
- Feature extraction is performed using TF-IDF / BoW
- Data is passed to ML and DL models for classification
- Classified articles are then summarized using LLMs
- End-to-end pipeline ensures automation and efficiency

Data Preprocessing & Feature Extraction for Text

Text Cleaning

Lowercasing, punctuation removal, stop-word filtering, and lemmatisation applied to standardise raw article text.

Bag of Words (BoW)

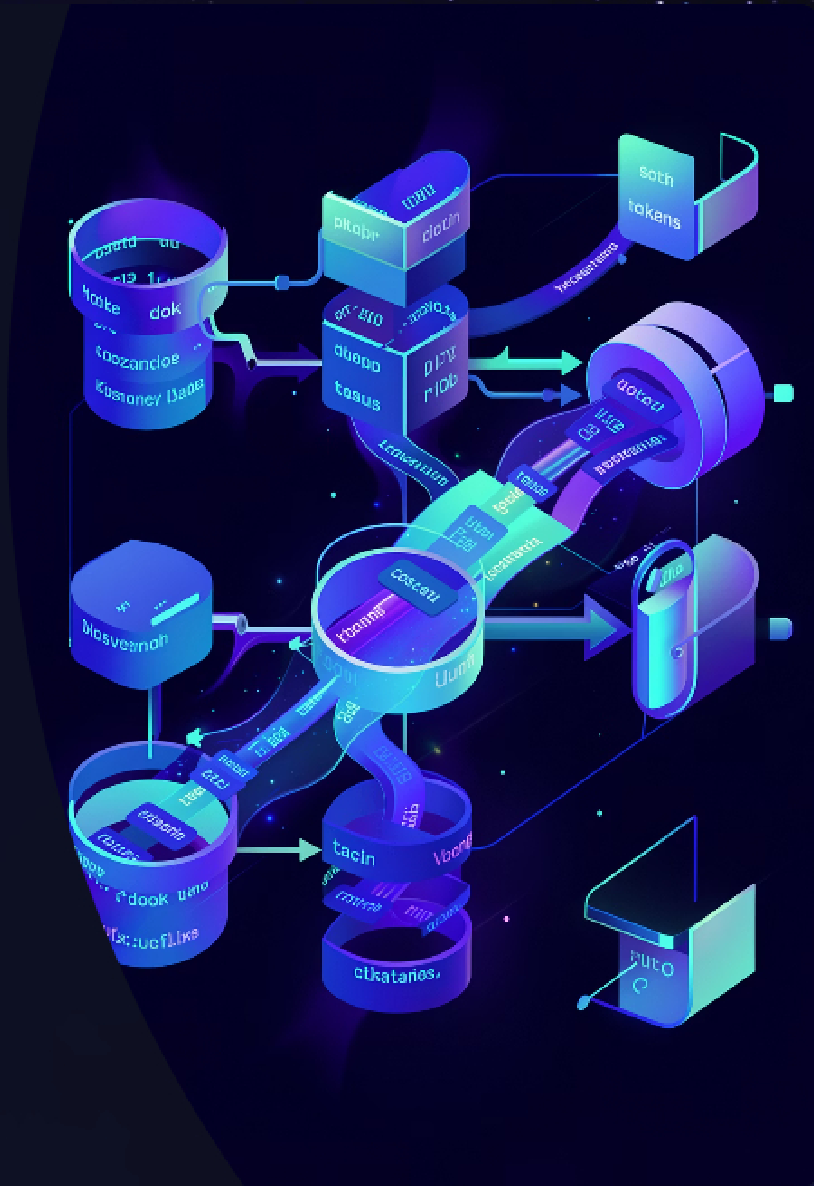
Represents text as word frequency counts. Simple, interpretable, but loses contextual word order information.

TF-IDF

Weighs term frequency against document frequency — surfaces **discriminative keywords** and reduces noise from common words.

Word Embeddings

Dense vector representations (GloVe/Word2Vec) used as input to deep learning models, capturing semantic similarity.



Models Used

Classical Machine Learning

→ **Naïve Bayes (NB)**

Probabilistic baseline; fast and effective for text.

→ **Logistic Regression (LR)**

Strong linear classifier with TF-IDF features.

→ **Support Vector Machine (SVM)**

Best-performing classical model with high-dimensional text vectors.

→ **Random Forest (RF)**

Ensemble method; robust but computationally heavier.

Deep Learning Models

→ **LSTM**

Long Short-Term Memory; captures long-range text dependencies.

→ **BiLSTM + Attention Layer**

Bidirectional context with attention weighting on key tokens — most expressive DL model.

ⓘ LLM integration (e.g. Llama / Gemini) used for abstractive summarisation post-classification.

Results & Model Comparison

Model	Accuracy (%)	F1-Score (%)
Naïve Bayes (BoW)	97.64	97.62
Logistic Reg. (BoW)	97.4	97.39
SVM (BoW)	**97.40**	**95.39**
Random Forest (BoW)	95.75	95.76
Naive Bayes (TF-IDF)	95.751	95.66
Logistic Reg (TF-IDF)	97.64	98.34
SVM ★ (TF-IDF)	**98.34**	**96.224**
Random Forest (TF-IDF)	96.22	91.2
LSTM	90.85	94
BiLSTM + Attention Layer	95.31	95.1

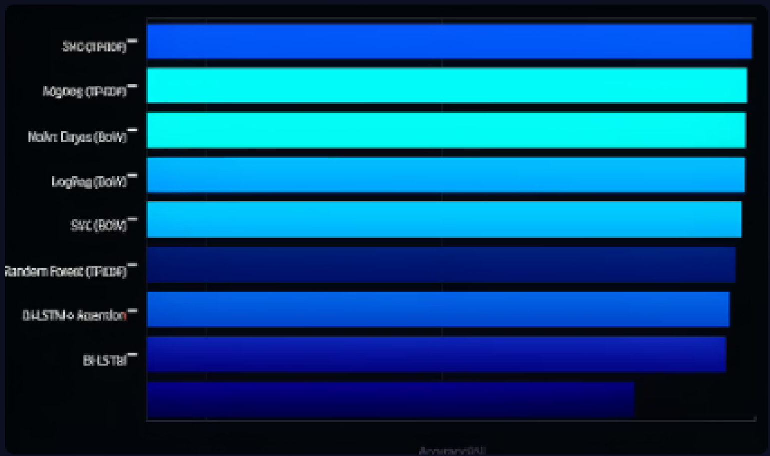
🏆 Best Overall

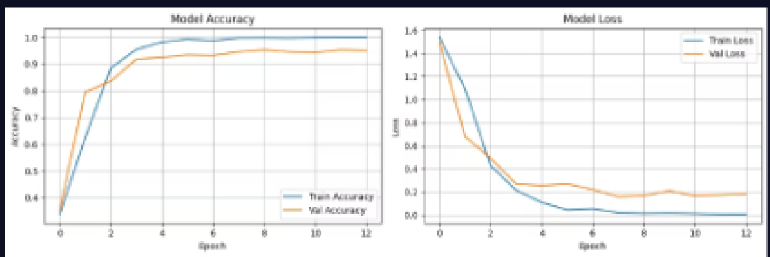
SVM + TF-IDF achieved 98.34% accuracy — outperforming all deep learning models on this dataset.

📊 DL Insight

BiLSTM with Attention is the strongest deep learning model at **95.31%**, benefiting from bidirectional context.

Graphs & Key Observations





Key Observations

TF-IDF dominates for classical ML

Sparse high-dimensional features give SVM a decisive edge over BoW representations.

DL models converge stably

Training loss curves for LSTM, and BiLSTM showed smooth convergence with no significant overfitting after dropout regularisation.

Attention improves DL accuracy

Adding attention to BiLSTM improved performance by ~5.3% over vanilla LSTM, confirming the value of token-level focus.

Contribution and Limitations

Contributions

Poposed an end-to-end pipeline for real-time news classification and summarization

Integration of Machine Learning, Deep Learning, and LLMs into a unified framework

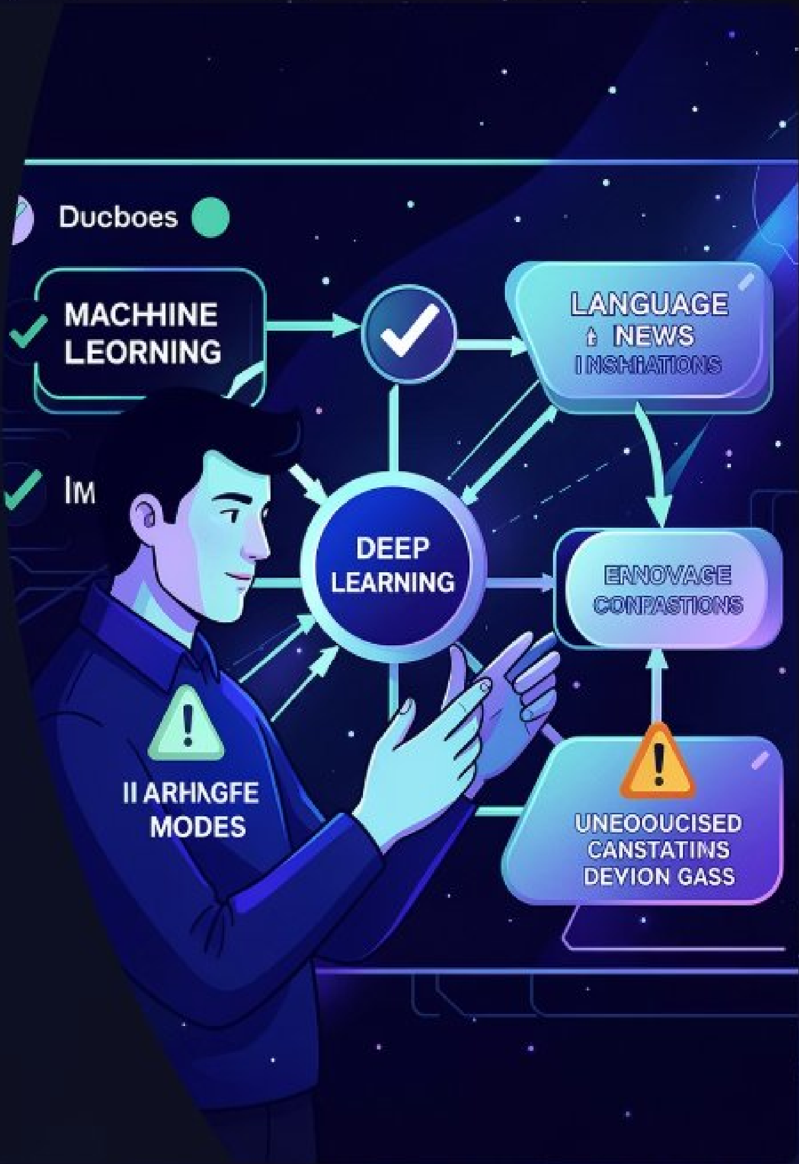
Implemented attention-based BiLSTM for improved contextual understanding

Compared BoW and TF-IDF feature extraction techniques

Demonstrated effectiveness of SVM + TF-IDF for news classification

Limitations

- ⚠ Due to time contraints, total integration of all components into a single framework was beyond reach
- ⚠ No human-in-the-loop feedback for continuous improvement
- ⚠ Reliance on a single extraction API (may affect reliability if API doesn't perform as expected)
- ⚠ System not fully deployed as a unified real-time pipeline
- ⚠ BBC dataset maybe simple and models classification ability on real data may get affected .



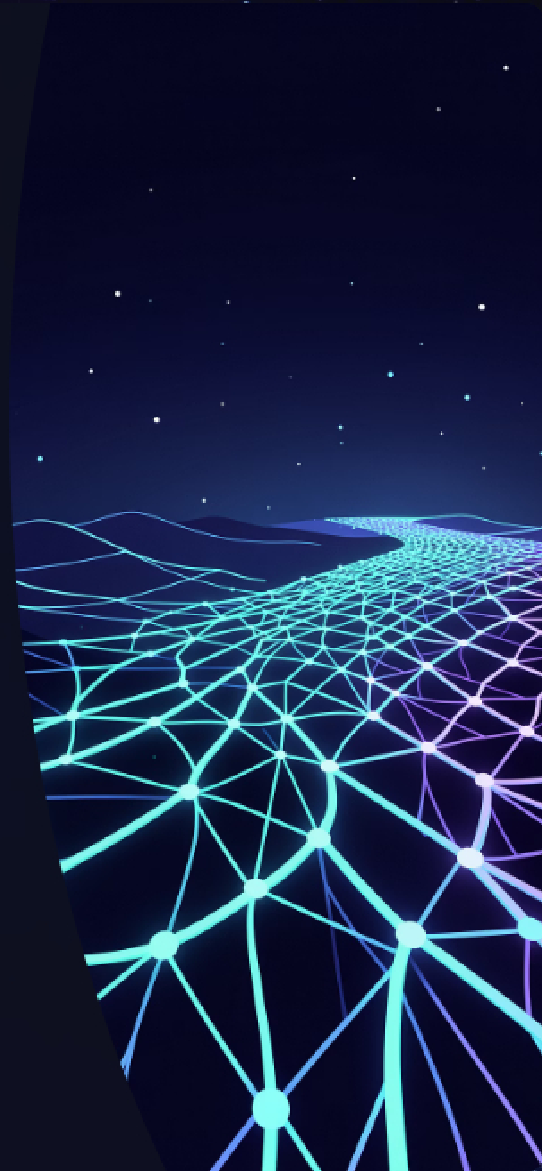
Conclusion, Limitations & Future Scope

✅ Conclusion

- SVM + TF-IDF achieved **98.34% accuracy** — the best result.
- BiLSTM + Attention led deep learning models at 95.6%.
- LLM summarisation produced coherent, abstractive summaries.
- TF-IDF outperformed BOW and advance word embeddings for this data

🚀 Future Scope

- Extend to **multilingual news**, fine-tune transformer models (BERT, GPT)
- Incorporate human-in-the-loop feedback for continuous learning and correction
- Implement intelligent agent-based extraction using multiple tools (e.g., Newspaper3k + APIs)
- Build a fully integrated end-to-end system (fetching, classification, summarization)



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